

Name:

BUID:

Lab 12: Type Inference

1. Determine the principal type of `fun x -> fun y -> fun z -> x (1 + y z)` using the process described in lecture (i.e., constraint-based inference, unification, and generalization).
2. Implement the function `val normalize : ty_scheme -> ty_scheme` so that `normalize ty` replaces the bound variables of `ty` with the `'a`, `'b`, `'c`, ... in order of appearance of in `ty`. That is, the following assertion should hold.

```
let _ = assert
  (normalize (["bar"; "foo"], TFun (TParam "foo", TFun (TParam "bar", TParam "foo"))))
    = (["a"; "b"], TFun (TParam "a", TFun (TParam "b", TParam "a"))))
```

You may assume that a type has fewer than 26 bound type parameters (you may also assume this for mini-project 3).

- ☐ Finished
- ☐ Started
- ☐ Did not start

Write your solutions below and on the back on this page.

Continue your work here.